

AAROGYAM (MEDADHERE)

Comprehensive Multi-Agent Architecture & System Manifest

Generated: September 14, 2026	Backend Stack: Django 5.x · DRF · Neon PostgreSQL · Celery
Scope: 26 Backend Agents · 14 Frontend UI Agents · 9 Celery Jobs	Frontend Stack: React 19 · Vite · Tailwind · WebSocket

1. System Architecture & Two-Tier Multi-Agent Model

The MedAdhere platform operates on a robust, decoupled **Two-Tier Agent System**. The backend domain services act as autonomous, single-responsibility agents communicating exclusively through typed HandoverPayload event contracts via the central AgentOrchestrator. On the frontend, client-side agents manage local caching, offline sync, and real-time reactive streams coordinated by the master FrontendOrchestrator.

2. Core Backend Domain Agents (Phases 1–12)

Agent Name	Phase	Domain & Primary Responsibilities	Status
AuthAgent	Phase 1	User registration, JWT token generation/rotation, session revocation, MFA (TOTP), and biometric authentication.	Active & Operational
PatientAgent	Phase 1	Patient profile lifecycle, chronic medical conditions, emergency contacts, onboarding, and consent tracking.	Active & Operational
CaregiverAgent	Phase 3	Caregiver-patient linking, permission delegation, emergency missed-dose escalation alerts, and adherence views.	Active & Operational
MedicationAgent	Phase 4	Prescription authoring, drug master database, dosage frequency, and rolling 7-day reminder slot generator.	Active & Operational
ReminderAgent	Phase 4	Due reminder dispatch, snooze window calculation (10–60 min), late dose grace tracking, missed dose tagging.	Active & Operational
AdherenceAgent	Phase 5	Multi-channel dose confirmation (App, Smart Pillbox, Caregiver), streak calculations, adherence rate metrics.	Active & Operational
NotificationAgent	Phase 7	Multi-channel dispatcher: Web push (FCM), SendGrid emails, Twilio SMS/WhatsApp, and IVR automated voice calls.	Active & Operational
IoTAgent	Phase 9	Smart Pill Dispenser pairing, device telemetry & heartbeat monitoring, real-time remote dispense via WebSocket / MQTT.	Active & Operational
AIAgent	Phase 8	XGBoost adherence risk inference, SHAP explanations, risk tiering (LOW/MED/HIGH), personalized behavioral nudges.	Active & Operational
SubscriptionAgent	Phase 2	Plan subscription tiers (Free, Pro, Family, Clinic), Razorpay webhook processing, feature gate enforcement.	Active & Operational
StoreAgent	Phase 10	E-commerce hardware store for Smart Pill Dispensers, order inventory management, hardware serial association.	Active & Operational
AuditAgent	Phase 11	HIPAA-compliant immutable audit logs, sensitive data access tracking, PHI access surveillance.	Active & Operational
AdminAgent	Phase 12	Super-admin dashboard analytics, system health metrics, tenant-wide user administration.	Active & Operational

3. Extension Backend Agents (Phases 13–28)

Agent Name	Phase	Domain & Primary Responsibilities	Status
PharmacyAgent	Phase 13	Auto-refill detection when compartment pills drop below threshold; pharmacy order dispatch.	Active & Operational
DoctorAgent	Phase 14	Doctor portal, teleconsultation room, WebRTC audio/video calls, digital prescription generation.	Active & Operational

WhatsAppBotAgent	Phase 15	Interactive conversational chatbot enabling feature phone users to log doses without smartphone apps.	Configured (Twilio)
DrugInteractionAgent	Phase 16	Live cross-check of newly prescribed medications against active patient drugs for adverse interactions.	Active & Operational
FamilyAgent	Phase 17	Multi-patient family accounts, profile switching using the X-Patient-Context header.	Active & Operational
FHIRAgent	Phase 18	HL7 / FHIR electronic health record (EHR) import and export for hospital integration.	Configured (EHR)
VitalsAgent	Phase 19	Vital signs ingestion (Blood Pressure, Blood Glucose, SpO2, Heart Rate, Temperature) and analytics.	Active & Operational
GamificationAgent	Phase 20	Adherence motivational nudges, streak badges, milestone achievements, point rewards.	Active & Operational
PharmacovigAgent	Phase 21	Adverse drug reaction (ADR) reporting, patient side-effect logging, clinical pharmacovigilance reports.	Active & Operational
InsuranceReportsAgent	Phase 24	Generates tamper-verified PDF adherence summary reports with cryptographic hashes for insurance discounts.	Active & Operational
GeofenceAgent	Phase 26	GPS location-aware smart reminders (remind patient if exiting home without their medications).	Active & Operational
ABHAAgent	Phase 27	Ayushman Bharat Digital Mission (ABDM) integration, ABHA ID verification, and health record linking.	Active & Operational
TenantAgent	Phase 28	Multi-tenant clinic isolation, white-label clinic portals, organizational role-based scoping.	Active & Operational

4. Frontend UI Agents (frontend/src/agents/)

Client-side singleton agents communicate with backend endpoints and handle reactive state via the master **FrontendOrchestrator**:

Frontend Agent	File Reference	Responsibilities & Capabilities	Status
FrontendOrchestrator	orchestrator.js	Coordinates cross-agent event bus (ORC_EVENTS), tabSync broadcast channel, and offline dose queues.	Active & Operational
adherenceAgent	adherence.agent.js	Fetches daily dose timeline, handles dose logging, snooze triggers, and dispense-now commands.	Active & Operational
aiAgent	ai.agent.js	Retrieves real-time adherence risk scores, SHAP explanations, insights, and doctor recommendations.	Active & Operational
authAgent	auth.agent.js	Manages user authentication, token refresh, biometric/MFA auth, and session teardown.	Active & Operational
caregiverAgent	caregiver.agent.js	Patient linking, caregiver alerts, emergency contact delegation, and patient adherence tracking.	Active & Operational
doctorAgent	doctor.agent.js	Prescription management, teleconsultation room, doctor-patient chat, and patient list queries.	Active & Operational
iotAgent	iot.agent.js	Hardware provisioning, dispenser LED/buzzer testing, WiFi reconfig, compartment mapping.	Active & Operational
familyAgent	family.agent.js	Switches active patient context, fetches dependent profiles, manages family member links.	Active & Operational
vitalsAgent	vitals.agent.js	Records vital signs metrics, queries historical vitals graphs, logs blood pressure and glucose.	Active & Operational
gamificationAgent	gamification.agent.js	Fetches adherence points, active streaks, unlocked badges, and achievement leaderboards.	Active & Operational
geofenceAgent	geofence.agent.js	Maintains home/work geofence radius and checks location triggers for dose reminders.	Active & Operational
pharmacyAgent	pharmacy.agent.js	Checks pill refill statuses and dispatches medication refill orders.	Active & Operational
prescriptionAgent	prescription.agent.js	Performs CRUD on active prescriptions and configures dose timing windows.	Active & Operational

userAgent	user.agent.js	Manages user settings, notification preferences, and profile updates.	Active & Operational
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5. Celery Automated Background Agents (config/celery.py)

Automated background cron agents that execute periodic business workflows via Celery Beat:

Task Identifier	Cron Schedule	Task Function & Responsibility	Status
dispatch-due-reminders	Every 2 mins	Pushes due reminder notifications (Push/SMS/WhatsApp) to patients for active dose windows.	Active & Operational
mark-missed-reminders	Every 5 mins	Scans unacknowledged dose windows past grace period, marks MISSED, and alerts caregivers.	Active & Operational
device-heartbeat-monitor	Every 5 mins	Checks IoT dispenser heartbeat timestamps. Flags devices missing >10 mins as offline.	Active & Operational
fill-reminder-window	Hourly (:00)	Generates ReminderJob records for all active patient prescriptions across rolling 7 days.	Active & Operational
purge-expired-sessions	Hourly (:00)	Deletes revoked JWT refresh tokens, blacklisted JTIs, and expired browser sessions.	Active & Operational
refill-alerts	Daily (09:00 UTC)	Scans remaining pill counts across all compartments; triggers auto-refill or alerts patient.	Active & Operational
subscription-expiry-check	Daily (01:00 UTC)	Checks active subscription ends. Demotes expired accounts to Free tier.	Active & Operational
nightly-risk-scoring	Daily (02:00 UTC)	Executes AI Engine batch XGBoost inference across all patients and updates risk scores.	Active & Operational
weekly-adherence-reports	Mondays (06:00 UTC)	Computes 7-day adherence averages and dispatches summaries to doctors and caregivers.	Active & Operational

Total Ecosystem Summary: 26 Backend Services · 14 Frontend UI Agents · 9 Celery Cron Agents.

Operational Guarantee: Hardware-Optional Zero-Dependency Design ensures patient adherence tracking operates via manual mobile tap, smart pillbox sensor, or caregiver proxy without data gaps.